

# SADINI PATHIRANA

## SOFTWARE ENGINEER

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[sadinipathirana](https://github.com/sadinipathirana)

### PROFILE

A motivated, hardworking, and responsible graduate with practical experience in software development and artificial intelligence. With a strong analytical and problem-solving mindset, I am eager to leverage my technical expertise to design, develop, and optimize intelligent systems.

### TECHNICAL SKILLS

**Programming Languages** – Java, Python

**Web Development** – HTML, CSS, JavaScript

**Frameworks** - MERN

**Databases** – MySQL, MongoDB, PostgreSQL

**Operating Systems** – Linux (Ubuntu)

**Cloud Platforms** – AWS

**DevOps & Tools** – Docker

### WORK EXPERIENCE

**Software Engineer Intern**

**2024 March- 2024 Aug**

*6IXSENSES Digital Transformation Inc, Canada*

- Actively contributed to Payroll System, developed and optimized server-side functionalities using Node.js and Express.js. Implemented RESTful APIs for automated salary sheet generation.
- Managed payroll data with MongoDB, ensuring data integrity and security while designing and maintaining schemas for employee and salary records.
- Designed and developed interactive payroll management interfaces with Next.js and styled them using Sass for a visually appealing design.

### AI/ML PROJECTS

**TourAI** | Final Year Research Project

Developed an AI-powered tourism solution to enhance revenue and traveler experiences in Sri Lanka. The system integrated fake review detection, sentiment analysis, and place recommendation to improve reliability and personalization. Understanding tourist preferences through sentiment analysis and recommending personalized travel places according to the weather and the user budget real-time insights can drive better resource allocation, marketing strategies, and service improvements.

- Technologies: Machine Learning (ML), Python, Graph Convolutional Networks (GCN)

**AI Chat Application**

A single page chat application with a React frontend and FastAPI backend. It uses Google Gemini via LangChain for conversation AI, with SQLite for chat history and session management.

- Technologies: FastAPI, Python, LangChain (Gemini), SQLite, React, TailwindCSS, Axios

**DocuChat AI**

An offline AI-powered chatbot that can read and answer questions from PDF documents. The system processes uploaded files, extracts relevant information, and enables interactive question-answering, allowing users to quickly retrieve insights from documents while ensuring data privacy.

- Technologies: Python, Flask, LangChain, GPT4All, FAISS, HuggingFace Sentence Transformers

## FULL STACK PROJECTS

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### Blockchain Based Vehicle Registration & Ownership Management System | Software Group Project

The Vehicle Registration and Ownership Management System is an application that facilitates digitizing the system of registering and transferring vehicle ownership with the use of Blockchain Technology. It enables to the conversion of physical assets into digital assets. Users who want to sell or buy vehicles in a secure manner can use this application.

- Technologies: Indy Blockchain, Django, Bootstrap, HTML, CSS, JavaScript, Docker Compose, Docker

### IoT-Based Smart Lighting System | Group Project

A solution to energy-efficient illumination using ESP8266 and Arduino IDE (C++). It automates lighting up of rooms depending on ambient conditions while allowing remote control using an app on mobile phones for the utmost convenience to users.

- Role: Embedded Systems Developer

## QA PROJECTS

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### Book Management System | Group Project

A book management system using a Spring Boot REST API to streamline book data management. the system is tested with Cucumber for behavior-driven development and Cypress for end-to-end testing to ensure reliability and functionality. The integration of these testing frameworks enhances test coverage, ensuring accurate and efficient management of book records.

- Technologies: Cypress, Cucumber

## EDUCATION

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### B.Sc. (Hons) in Information Technology

2025

University of Moratuwa, Sri Lanka

- CGPA : 3.3 (Best GPA: 3.71 (L2S2))

## CERTIFICATES & ACHIEVEMENTS

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Web Design for Beginners - Certificate | [Link](#)

Python for Beginners - Certificate | [Link](#)

Full stack generative and Agentic AI with python – Certificate | [Link](#)

## VOLENTEER EXPERIENCE

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### Rotaract Club of University of Moratuwa

- "Are you ready? 2023" Official Career Fair UoM - Company Coordination
- Data Strom 4.0 - Organizing Committee Member
- Rotaract Alumni - Partnership and Pitching Team Member

### IEEE Student Branch - University of Moratuwa

- Rise Up Mora 2023 - Publicity Committee Member

## SOFT SKILLS

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Team Work, Time Management, Quick Learner, Communication Skills, Adaptability, Leadership

## REFEREES

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### Dr. A. L. A. R. R. Tanuja

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